P02 – ARPAV temperature trends in Veneto region

Field: Meteorology and Climate

Scientific question. Is there evidence for a systematic trend in temperature at meteorological stations in Veneto?

Data. Use the ARPAV open-data archive "Principali variabili meteorologiche", distributed as ZIP files containing CSV data for individual stations:

- https://www.arpa.veneto.it/dati-ambientali/open-data/clima/principali-variabili-meteorologiche
- Alternative CKAN page: https://opendata.arpa.veneto.it/dataset/principali-variabili-meteorologiche

Use 3 stations (for example, Auronzo di Cadore, Castelfranco Veneto, and Porto Tolle).

Models.

- Model 1: constant mean annual temperature for each station.
- Model 2: linear trend in time for each station,

$$T_{s,t} = T_s^0 + \dot{T}_s t + \epsilon_{s,t}$$

where s labels the station, and ϵ is a Gaussian noise with variance σ_s^2.

Required output.

1. Plot monthly and annual averages of minimum, mean and maximum temperature.
2. Infer station-specific parameters and their credible intervals.
3. Compare constant and linear-trend models.
4. Produce posterior predictions for the next 10 years for the annual mean temperature.

References.

- ARPAV, Open Data Meteo e Clima: https://www.arpa.veneto.it/dati-ambientali